PDM

POSITIVE DISPLACEMENT MOTOR WITH SLEEVE STABILIZER

Enedril

DRILLING

GENERAL DESCRIPTION

POSITIVE DISPLACEMENT MUD MOTOR (PDM) FOR DIRECTIONAL AND PERFORMANCE DRILLING, FROM 3-1/8 TO 9-5/8 IN. IN STOCK. THE MOTOR IS BUILT FROM A POWER SECTION (ROTOR AND STATOR), AN ADJUSTABLE OR FIXED BENT HOUSING, A DRIVESHAFT ASSEMBLY, A SEALED OR MUD-LUBRICATED BEARING ASSEMBLY AND THE BIT BOX. THE BEARING HOUSING CARRIES A THREAD FOR A SCREW-ON SLEEVE STABILIZER (STRAIGHT OR SPIRAL BLADE) OR OFFSET PAD, USED TO TUNE THE BUILD RATE AND TO REDUCE LATERAL VIBRATION. SPEED IS PROPORTIONAL TO FLOW, TORQUE TO DIFFERENTIAL PRESSURE.

TECHNICAL CHARACTERISTICS

Power section OD (in)	Lobes	Stages	Max. flow (gpm)	Max. bit speed (rpm)	Std. max diff. press. (psi)	Std. torque (lb-ft)	High-power diff. press. (psi)	High-power torque (lb-ft)
3-3/8	7/8	3.0	120	188	460	782	690	1,170
3-3/4	6/7	3.0	200	160	440	1,314	660	1,970
4-3/4	7/8	2.6	300	79	390	3,500	590	5,250
5	6/7	8.0	350	302	1,215	3,500	1,820	5,250
6-1/4	7/8	2.9	500	82	425	6,277	640	9,420
6-1/2	8/9	4.0	500	145	605	4,674	910	7,010
6-3/4	7/8	5.7	600	150	845	8,204	1,270	12,310
7	7/8	7.5	600	186	1,120	8,508	1,680	12,760
7-3/4	7/8	4.0	900	149	570	8,333	860	12,500
8	6/7	5.0	900	159	715	9,618	1,070	14,430
9-5/8	6/7	6.0	1,200	155	880	15,775	1,320	23,660
11-1/4	6/7	5.6	1,800	162	820	19,377	1,230	29,070

One configuration per size: the highest-torque standard power section listed. Other lobe and stage combinations are available. High-power = enhanced elastomer power section (about 50% higher differential pressure). Values per manufacturer motor tables, for reference.

DESIGN FEATURES

- **Modular motor:** power section, bent housing, driveshaft, bearing assembly and bit box can be selected to suit the job.
- **Adjustable bend:** field-adjustable housing (2.38, 3 or 4 degrees) for a wide range of build rates; fixed housings on request.
- **Sleeve stabilizer option:** threaded bearing housing takes a screw-on straight or spiral blade sleeve, or an offset pad.
- **Bearing assembly:** oil sealed (100% of the flow to the bit) or mud lubricated.
- **Driveshaft:** conventional or high-torque version for the strongest power sections.
- **Dump sub and rotor catch:** bypass while tripping; retrieval of the motor after a housing failure.

PERFORMANCE PARAMETERS

Speed	Proportional to flow rate
Torque	Proportional to differential pressure, independent of flow
More torque	More stages, enhanced elastomer or even-wall stator
Stall	Standpipe pressure rises, penetration stops
If stalled	Stop rotation, slow or stop pumps, release torque slowly, lift off bottom

HOW TO ORDER

Please specify: 1. motor OD and power section (lobes, stages, standard or high-power) 2. bearing type 3. housing angle 4. sleeve stabilizer type and size, or offset pad 5. top sub / dump sub and bit box connection 6. fluid, temperature and flow range.